

Office and PDF Tools

This document describes ragent’s **office document tools** – the MS Office (`office_read`, `office_write`, `office_info`) and LibreOffice/OpenDocument (`libre_read`, `libre_write`, `libre_info`) families – plus the two PDF tools (`pdf_read`, `pdf_write`). It covers formats, JSON input schemas, worked examples, permissions, checkpointing, and configuration options.

All parsing and writing is done **in-process with pure-Rust crates**. There is no `soffice`/LibreOffice binary, no Microsoft Word, and no external converter to install: the “libre” tools read OpenDocument ZIP/XML directly, and the PDF tools use in-process extraction/writer crates. A workspace-wide `grep` for `soffice` returns zero matches.

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1. Tool inventory

Eight tools are registered by `crates/ragent-tools-extended/src/lib.rs` (lines 430-437) in this order:

Tool name	Family	Purpose	Permission
<code>pdf_read</code>	PDF	Extract text or metadata from PDF files	<code>file:read</code>
<code>pdf_write</code>	PDF	Create new PDF documents	<code>file:write</code>
<code>office_read</code>	MS Office	Read Word/Excel/PowerPoint (OOXML) content	<code>file:read</code>
<code>office_write</code>	MS Office	Create Word/Excel/PowerPoint (OOXML) files	<code>file:write</code>
<code>office_info</code>	MS Office	Return JSON metadata for OOXML files	<code>file:read</code>
<code>libre_read</code>	LibreOffice	Read ODT/ODS/ODP content	<code>file:read</code>

Tool name	Family	Purpose	Permission
libre_write	LibreOffice	Create ODT/ODS/ODP files	file:write
libre_info	LibreOffice	Return structure summary for ODF files	file:read

All eight are implemented in `crates/ragent-tools-extended/src/` (modules `office_read.rs`, `office_write.rs`, `office_info.rs`, `libreoffice_read.rs`, `libreoffice_write.rs`, `libreoffice_info.rs`, `pdf_read.rs`, `pdf_write.rs`, plus shared helpers `office_common.rs` and `libreoffice_common.rs`).

2. Enabling the tools

The office/PDF family is **hidden from the LLM by default**. The `tool_visibility.office` switch (a bool, default `false`) controls whether the eight tools are advertised in `ToolRegistry::definitions()`; two consequences follow:

- When the switch is `false`, all tools in the family are suppressed from the tool definitions the model sees, so the model will not discover or call them on its own.
- Tools remain **registered and executable regardless of visibility** – a call by name still runs. Visibility only affects advertising, not executability.

Two ways to enable:

```
# 1. TUI slash command (persists to config)
/tools office on

// 2. ragent.json
{
  "tool_visibility": {
    "office": true
  }
}
```

Other `/tools` switches (`github`, `gitlab`, `teams`, `agents`, `plan`, `codeindex`, `masterfetch`, `browser`) behave the same way; `/tools office` without an argument reports the current state. The `/tools` handler persists the change with `Config::save()` (which invalidates the load cache, so the change is immediately visible to subsequent loads in the same process) and reports `[ok] 'office' visibility is now **true**`. A save failure is reported with a `[warn]` line and an `(unsaved)` status hint.

For reference, the default visibility switches are:

Switch	Default
<code>office</code>	<code>false</code> (hidden)
<code>github</code>	<code>false</code> (hidden)
<code>gitlab</code>	<code>false</code> (hidden)
<code>teams</code>	<code>false</code> (hidden)
<code>agents</code>	<code>false</code> (hidden)
<code>plan</code>	<code>false</code> (hidden)

Switch	Default
<code>codeindex</code>	<code>true</code>
<code>masterfetch</code>	<code>true</code>
<code>browser</code>	<code>true</code>
<code>finance</code>	<code>true</code>

3. Format matrix

Family	Formats supported	Legacy formats
<code>office_*</code>	<code>.docx</code> (Word), <code>.xlsx</code> (Excel), <code>.pptx</code> (PowerPoint) – OOXML only	<code>.doc</code> , <code>.xls</code> , <code>.ppt</code> are rejected with an actionable error
<code>libre_*</code>	<code>.odt</code> (text), <code>.ods</code> (spreadsheet), <code>.odp</code> (presentation) – OpenDocument	non-ODF extensions rejected
<code>pdf_*</code>	<code>.pdf</code>	–

Legacy binary Office formats (`.doc`, `.xls`, `.ppt`) are **rejected** with a message directing conversion: Please convert to the modern OOXML format (`.docx`, `.xlsx`, `.pptx`). Files with no recognisable extension are rejected the same way. Convert legacy documents with LibreOffice, Word, or soffice `--convert-to docx` before using these tools.

4. Tool reference

4.1 `office_read`

Reads content from Word (`.docx`), Excel (`.xlsx`), or PowerPoint (`.pptx`) files.

Parameter	Type	Required	Notes
<code>path</code>	string	yes	Path to the file
<code>sheet</code>	string	no	Excel sheet name or index (0-based)
<code>range</code>	string	no	Excel cell range, e.g. <code>A1:D10</code> (1-based cells)
<code>slide</code>	integer	no	PowerPoint slide number (1-based)
<code>format</code>	enum	no	<code>text</code> , <code>markdown</code> (default), or <code>json</code>

The tool runs off the main thread (`spawn_blocking`), renders the selected portion, and truncates large output (Section 5). Markdown is the default presentation: paragraphs become text, tables become markdown tables, headings become `#`-prefixed lines.

4.2 office_write

Creates or overwrites OOXML files. Required: `path` + `content`. Optional `type` (`docx`, `xlsx`, or `pptx` – auto-detected from the file extension when omitted) and `title` (`docx` only). Parent directories are created automatically.

The `content` payload is format-specific JSON:

- **docx** – an array of blocks: `{"type": "paragraph" | "heading" | "bullet_list" | "code_block", "text": ..., "level": ..., "items": [...], "style": ...}` (only the fields relevant to the block type are required).
- **xlsx** – `{"sheets": [{"name": "Sheet1", "rows": [{"a", "b"}, [{"1", "2"}]]]}`.
- **pptx** – `{"slides": [{"title": ..., "body": ..., "notes": ...}, ...]}`; a direct top-level array of slide objects is also accepted.

The writer tolerates several payload shapes for `docx` (3 variants) and `pptx` (5 variants) plus a top-level-key fallback, so slightly different JSON structures still succeed.

4.3 office_info

Takes only `path` and returns JSON metadata: title, author, creation date, page/slide/sheet counts, word count (`docx`), sheet names (`xlsx`), and slide titles (`pptx`). Legacy `.doc/.xls/.ppt` inputs are rejected.

4.4 libre_read

Reads `.odt`, `.ods`, and `.odp` content.

Parameter	Type	Required	Notes
<code>path</code>	string	yes	Path to the OpenDocument file
<code>sheet</code>	string	no	ODS only – sheet name or index
<code>range</code>	string	no	ODS only – cell range, e.g. <code>A1:D10</code>
<code>slide</code>	integer	no	ODP only – slide number (1-based)
<code>format</code>	enum	no	<code>text</code> , <code>markdown</code> (default), <code>json</code>

Implementation: ODS is read with the `calamine` crate; ODT and ODP are parsed by extracting text from the ODF ZIP's XML (via `quick-xml`).

4.5 libre_write

Creates OpenDocument files. Only `path` is required.

Parameter	Type	Notes
<code>path</code>	string	Destination file (<code>.ods</code> , <code>.odt</code> , or <code>.odp</code>)
<code>rows</code>	2-D string array	ODS content; header row first
<code>content</code>	array or text	ODT/ODP block array or plain text
<code>sheet_name</code>	string	ODS sheet name
<code>title</code>	string	Document title
<code>author</code>	string	Document author

Block payloads for ODT/ODP use `{"type": "paragraph" | "heading" | "bullet_list" | "ordered_list" | "code_block", ...}`; plain text is also accepted. ODS is written with the `spreadsheet-ods` crate; ODT/ODP are written as ODF ZIP + XML directly. Note the asymmetry with `office_write`: only `path` is required here.

4.6 `libre_info`

Requires `path`; the extension must be `.ods`, `.odt`, or `.odp`. Returns sheet list with dimensions (ODS), word/paragraph counts (ODT), or slide count (ODP).

4.7 `pdf_read`

Extracts text from PDF files in-process.

Parameter	Type	Required	Notes
<code>path</code>	string	yes	Path to the PDF
<code>start_page</code>	integer	no	1-based, inclusive
<code>end_page</code>	integer	no	1-based, inclusive
<code>format</code>	enum	no	<code>text</code> , <code>metadata</code> , or <code>json</code>

Page extraction works by extracting the whole document and filtering to the requested range, with an `lpdf` fast-path fallback. The `metadata` format returns `path/format/line-count` metadata; line counts reflect the (possibly truncated) text. Output is truncated at the shared 100 KB limit.

Encrypted PDFs are not supported: the vendored `pdf-extract` crate only decrypts through password-taking variants that `pdf_read` does not use, so an encrypted document errors out.

4.8 `pdf_write`

Creates a new PDF from a structured payload. Requires `path` + `content`; the `content` object has an optional `title` and an `elements` array whose items are `{type: "paragraph" | "heading" | "table" | "image", text, level, headers, rows, ...}` (`level` 1-3 for headings; `headers/rows` for tables). Layout is fixed A4 (210x297 mm) with 25 mm margins and built-in font sizes (11 pt body, 22 pt H1, 17 pt H2, 13 pt H3). Parent directories are created automatically.

5. Output formats and truncation

- Successful reads truncate at `MAX_OUTPUT_BYTES = 100 KiB` (`office_common.rs`). The truncation notice explicitly tells the model to narrow the result using `sheet/range` (Excel, ODS) or `slide` (PowerPoint, ODP) instead of reading whole large documents.
- `pdf_read`'s reported line count reflects the text *after* truncation.
- Write errors embed a truncated JSON snippet of the offending payload in the error message.
- There is no `offset/limit` pagination parameter on the office tools (unlike `MasterFetch`'s web fetches); pagination is by `sheet`, `range`, or `slide` only.

6. Permissions and checkpointing

- **Permission categories.** Read-side tools (`office_read`, `office_info`, `libre_read`, `libre_info`, `pdf_read`) require `file:read`; write-side tools (`office_write`, `libre_write`, `pdf_write`) require `file:write`. These flow through the standard configurable allow/deny/ask rule engine and the file-path guards – the same surface as the plain read/write tools. There are no office-specific permission rules.
 - **Goal-loop write detection.** `LOOP_WRITE_TOOLS` (in `crates/ragent-agent/src/session/loop_state.rs`) includes `office_write`, `libre_write`, and `pdf_write`. Inside a `/loop` run, any call to these three tools counts as a destructive write: it arms the pre-loop snapshot/checkpoint machinery (forced checkpoints gated on `spec.checkpoints` and `checkpoint_timeout_secs`) and participates in the loop change summary/diffstat.
 - **Auto-approval interplay.** `--yes` (alias `--no-prompt`) and YOLO mode auto-approve ordinary permission prompts, but forced loop checkpoints still fire for these write tools while loop checkpoints are enabled.
 - **Visibility vs executability.** With `tool_visibility.office = false` the tools are hidden from the model but a call by name still executes.
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7. Implementation notes

- All parsing and writing is pure Rust – no LibreOffice/soffice subprocess, no Word installation, no headless converters. The “libre” tools unpack the ODF ZIP and parse `content.xml/meta.xml` with `quick-xml`.
 - Underlying crates (from `crates/ragent-tools-extended/Cargo.toml`): `docx-rust` and `ooxmlsdk` (Word), `calamine` (Excel/ODS reading), `spreadsheet-ods` (ODS writing), `zip`, `quick-xml`, `printpdf` (PDF writing), `pdf-extract` (PDF reading, vendored at `vendor/pdf-extract`), `lopdf` (PDF fast path).
 - `pdf-extract` is a workspace-patched dependency: the root `Cargo.toml` declares `pdf-extract = "0.10"` and patches it to `vendor/pdf-extract`. Extraction is whole-document-in-memory (`extract_text_from_mem`); encrypted variants exist upstream but are unused.
 - Both `office_read` and `office_write` execute on the tokio blocking pool (`spawn_blocking`), so large document work does not stall the async runtime.
 - Office/PDF writes do not take their own file snapshots inside the tool; the snapshot/undo machinery that covers them is the session-level snapshot path and, inside goal loops, the forced-checkpoint machinery described in Section 6.
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8. Worked examples

The examples below show the JSON arguments an agent passes to each tool. In the TUI you would normally just ask for the outcome (“read the table from `report.xlsx`”); the model issues these calls itself.

8.1 Read a Word document as markdown

```
{ "path": "docs/report.docx", "format": "markdown" }
```

8.2 Read one Excel sheet and range

```
{ "path": "data/metrics.xlsx", "sheet": "Q3", "range": "A1:D40", "format": "json" }
```

8.3 Read a single PowerPoint slide

```
{ "path": "decks/launch.pptx", "slide": 7 }
```

8.4 Create a Word document

```
{
  "path": "out/summary.docx",
  "type": "docx",
  "title": "Sprint Summary",
  "content": [
    { "type": "heading", "text": "Summary", "level": 1 },
    { "type": "paragraph", "text": "All planned work landed this iteration." },
    { "type": "bullet_list", "items": ["auth rewrite", "SSE hardening", "bench regressions fixed"] }
  ]
}
```

8.5 Create a spreadsheet

```
{
  "path": "out/budget.xlsx",
  "type": "xlsx",
  "content": {
    "sheets": [
      { "name": "Budget", "rows": [["Item", "Cost"], ["Hosting", "120"], ["CI", "45"]] }
    ]
  }
}
```

8.6 Create a presentation

```
{
  "path": "out/kickoff.pptx",
  "content": {
    "slides": [
      { "title": "Roadmap", "body": "Q3 objectives", "notes": "keep to 10 minutes" },
      { "title": "Risks", "body": "quota and supply", "notes": "" }
    ]
  }
}
```

8.7 Inspect OOXML metadata

```
{ "path": "docs/spec.pdf" }
```

(office_info on a .docx returns title/author/creation date/word count; on .xlsx it adds sheet names; on .pptx slide titles.)

8.8 Read OpenDocument files

```
{ "path": "sheets/totals.ods", "sheet": "Summary", "range": "A1:C12" }
{ "path": "notes/plan.odt", "format": "text" }
{ "path": "decks/pitch.odp", "slide": 3 }
```

8.9 Write an OpenDocument spreadsheet

```
{
  "path": "sheets/plan.ods",
  "sheet_name": "Plan",
  "rows": [["Phase", "Owner"], ["Discovery", "alice"], ["Build", "bob"]],
  "title": "Delivery plan",
  "author": "ragent"
}
```

8.10 Create a PDF report

```
{
  "path": "reports/audit.pdf",
  "content": {
    "title": "Dependency audit",
    "elements": [
      { "type": "heading", "level": 1, "text": "Summary" },
      { "type": "paragraph", "text": "All direct dependencies pass cargo-deny." },
      { "type": "table",
        "headers": ["Crate", "Licence", "Status"],
        "rows": [["pdf-extract", "MIT", "ok"], ["printpdf", "MIT", "ok"]] }
    ]
  }
}
```

8.11 Enable the family and drive a document task

```
# one-time enable (or use /tools office on in the TUI)
# ragent.json: { "tool_visibility": { "office": true } }
ragent run --agent general "read assets/pdf/Profile.pdf and summarise the candidate's experience"
```

Sample files for manual testing live in assets/pdf/ (Profile.pdf, vibeSDLC.pdf) and assets/officedocs/ (testword1.docx).

9. Commands and configuration reference

TUI

Command	Effect
/tools	Show the tools/visibility help

Command	Effect
/tools office	Report whether the office family is currently visible
/tools office on/off	Toggle visibility and persist to config
/agents	Agent diagnostics include the Office family status

Configuration (ragent.json)

Key	Type	Default	Effect
tool_visibility.office_tool	bool	false	When false, all eight office/PDF tools are suppressed from ToolRegistry::definitions() (hidden from the LLM). Tools stay registered and executable regardless.

There are no office/pdf-specific keys beyond the visibility switch; all other behaviour (formats, payloads, limits) is fixed by the tool implementations.

Permissions

Category	Tools
file:read	office_read, office_info, libre_read, libre_info, pdf_read
file:write	office_write, libre_write, pdf_write

Configure allow/deny/ask rules for these categories in the permissions array of ragent.json, exactly as for the plain file tools.

10. Testing and sample files

- **Behavioural unit tests for the eight tools are currently absent** – there are no test_office_*, test_libre_*, or test_pdf_* tool-behaviour files under crates/ragent-tools-extended/tests/. Coverage that does exist is indirect: the visibility listing includes all eight names (crates/ragent-agent/tests/test_tui_definitions.rs), the TUI definitions() check includes office_read (crates/ragent-tui/tests/test_slash_commands.rs), and the step-log input summary for libre_read is locked in by crates/ragent-tui/tests/test_tool_display.rs.
- Sample asset files (assets/officedocs/testword1.docx, assets/pdf/Profile.pdf, assets/pdf/vibeSDLC.pdf) are informal manual fixtures; nothing in tests/ or examples/ references them, so they are not CI coverage.
- vendor/pdf-extract is a workspace-patched dependency used by pdf_read (and MasterFetch's PDF path); it extracts text fully in memory with no external binary.

11. Related documentation

- `docs/howtos/reactagent.md` – the core per-turn agent loop that executes these tools.
- `docs/howtos/loopprogramming.md` – goal loops; relevant for how `office_write/libre_write/pdf_write` participate in forced checkpoints.
- `TUI-QUICKSTART.md` – quick reference for slash commands including `/tools`.
- `crates/ragent-tools-extended/DOCS.md` – curated descriptions of the extended tool families.
- `SPEC.md` – full configuration schema including `tool_visibility`.